

PRODUCT REQUIREMENTS DOCUMENT

Android App Development using GenAI - Hasta-Kala Shop

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Project Namma-Platform	Program MindMatrix VTU Internship — Project #14	Version 1.0
Platform Android (Android Studio)	Status Draft — Pending Review	Date April 2026
Developer Rakshita A U	USN / College 1KS22CS113 K.S Institute of Technology	IDE Android Studio

1. Executive Summary

Hasta-Kala Shop is a simple Android-based micro-sales analytics application built to support artisans and small craft sellers in managing inventory and understanding sales trends.

Traditional artisans often create products without knowing which colors, designs, or items sell the most, leading to unsold inventory and wasted effort. This application provides a lightweight billing and analytics system that helps artisans log every sale, view best-selling items, track stock levels, and make informed production decisions.

The app combines simple digital management with GenAI-assisted insights to support self-employment and improve business growth.

2. Problem Statement

2.1 Context

Small-scale artisans making handmade products such as banana fiber bags, keychains, and crafts usually manage sales manually.

They face challenges such as:

- No proper inventory tracking
- Difficulty identifying best-selling products

- Unsold products leading to dead stock
- Lack of data to guide production decisions

2.2 Core Pain Points

For Artisans:

1. No record of frequently sold products.
2. Overstocking of low-demand designs.
3. Manual bookkeeping errors.
4. No alerts when stock is low.
5. No simple digital tool designed for micro-sellers.

2.3 Impact

Affected Group	Problem
Artisans	Cannot identify best sellers
Small Craft Sellers	Poor inventory decisions
Rural Entrepreneurs	Lack digital sales management
Customers	Popular items often go out of stock

3. Product Vision and Goals

3.1 Vision Statement

“Empowering artisans with simple digital tools and sales intelligence to make smarter production decisions.”

3.2 Goals

Goal ID	Goal	Category
G1	Enable quick sale logging	Core
G2	Show best-selling products	Analytics
G3	Provide low-stock alerts	Inventory
G4	Track income weekly/monthly	Business
G5	Use GenAI insights for trends	Innovation
G6	Create intuitive visual dashboard	UX

4. Target Users & Personas

Persona 1 – Rural Artisan

Name: Lakshmi

Occupation: Handcrafted Bag Maker

Needs:

- Easy billing system
- Sales trend insights
- Inventory support

Goal: Know what products to make more of.

Persona 2 – Micro Entrepreneur

Name: Ravi

Occupation: Craft Stall Owner

Needs:

- Track daily income
- View popular products
- Low stock notifications

Goal: Improve profits and reduce dead stock.

5. Functional Requirements

FR-01 Quick Bill

Users can log product sales quickly.

Features:

- Select item
- Select color/design
- Save sale

Acceptance Criteria

- Sale should save instantly.
- Quantity updates automatically.
- Easy fast-entry interface.

FR-02 Best Seller Dashboard

Display popular items through analytics.

Features:

- Pie chart of product sales
- Bar chart trends
- Most popular products section

Acceptance Criteria

- Chart updates after every sale.
 - Best seller ranking updates dynamically.
 - Clear visual representation.
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FR-03 Stock Alert

Shows low-stock notifications.

Example: “Only 2 Blue Bags Left — Time to Make More”

Acceptance Criteria

- Auto low stock alerts.
 - Threshold configurable.
 - Alerts visible on dashboard.
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FR-04 Income Log

Track total revenue.

Features:

- Daily sales total
- Weekly summary
- Monthly summary
- Filters by date

Acceptance Criteria

- This Week filter works.
 - This Month filter works.
 - Income totals calculate correctly.
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FR-05 GenAI Sales Insight (Bonus)

GenAI suggests production recommendations.

Examples:

- “Red Bags are trending.”
- “Increase production of keychains.”

Acceptance Criteria

- AI recommendations based on sales data.
 - Suggestions easy to understand.
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6. App Usage & User Flow

Primary Flow

1. User opens app.
2. Selects product and records sale.
3. Inventory updates.
4. Dashboard reflects updated best sellers.
5. Stock alert appears if quantity low.
6. User checks income log.
7. User uses insights to decide future production.

Flow Diagram

Login / Home
↓ Quick Bill
↓ Sales Recorded
↓ Analytics Dashboard
↓ Stock Alerts
↓ Income Insights

7. Technical Requirements

7.1 Technology Stack

Component	Technology
Platform	Android
Language	Java / Kotlin
IDE	Android Studio

Component	Technology
Database	Room Database
Charts	MPAndroidChart
UI	XML + ConstraintLayout
GenAI	Gemini API (Optional)

7.2 Database Schema (Sample)

Products Table:

- Product ID
- Product Name
- Color
- Quantity
- Price
- Units Sold

Sales Table:

- Sale ID
 - Product ID
 - Date
 - Quantity Sold
 - Revenue
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7.3 Technical Hints

As given in project brief:

Visualization

Use MPAndroidChart for:

- Pie Charts
- Bar Charts

Database

Use Room DB for:

- Sales history
- Inventory records

UI

Use icons for products for quick entry.

8. Non-Functional Requirements

Requirement	Target
App Response Time	<3 sec
Data Accuracy	100% correct sales logging
Reliability	No crashes
Usability	Beginner friendly
Offline Support	Core features available offline
Accessibility	Simple visual interface

9. UI/UX Requirements

Design Principles

Simple & Visual

- Icon based navigation
- Minimal text entry
- Easy for non-technical users

Analytics Focused

- Visual dashboards
- Graph-driven decisions

Clean Interface

- Product cards
 - Large buttons
 - Quick billing workflow
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Main Screens

Screen 1 — Quick Bill

Contains:

- Product icons

- Color selection
- Save sale button

Screen 2 — Best Seller Dashboard

Contains:

- Pie chart
- Bar chart
- Popular product cards

Screen 3 — Stock Alert Panel

Contains:

- Low stock warnings
- Restock suggestions

Screen 4 — Income Log

Contains:

- Weekly sales report
- Monthly sales report
- Filters

10. Success Criteria & Evaluation

Based on project requirements:

Criterion	Target
Best Seller chart updates after sale	Must Have
Income log filtering works	Must Have
Intuitive visual UI	Must Have
Stock alerts trigger correctly	Must Have
Analytics charts display correctly	Must Have
GenAI suggestion feature	Nice to Have

11. Impact Goals

Business Intelligence

Gives micro-entrepreneurs data power similar to larger retailers.

Waste Reduction

Reduces unsold goods and overproduction.

Growth Mindset

Encourages artisans to track business progress.

Self-Employment Support

Helps small sellers grow sustainably using technology.

12. Out of Scope (MVP)

Not included in version 1:

- Online payments
 - Full e-commerce marketplace
 - Customer accounts
 - Product delivery system
 - Advanced demand forecasting
 - Multi-store management
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13. Assumptions & Constraints

Assumptions

- Artisans can log each sale.
- Product data is maintained correctly.
- Smartphone available to user.
- App used for small-scale businesses.

Constraints

- Limited development time.
 - Android-only implementation.
 - Basic MVP features prioritized.
 - GenAI optional for demo scope.
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14. Suggested Milestones

Week	Deliverable
Week 1	Project setup + database design
Week 2	Quick Bill implementation
Week 3	Best Seller dashboard
Week 4	Stock alerts + Income log
Week 5	GenAI integration + testing
Week 6	Final polishing + demo prep

15. Future Enhancements

Possible upgrades:

- Voice-based billing in local languages
 - Demand forecasting using AI
 - Customer ordering feature
 - Marketplace integration
 - Multi-artisan collaboration platform
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Role	Name	Signature / Date
Product Owner / Developer	RAKSHITA A U	1KS22CS113
Technical Lead / Mentor		
Internship Coordinator		